

Robin Karp algorithm

Hand calculation

$$T \leftarrow cdba \ (3421)$$

$$P \leftarrow dba \ (421)$$

$$q=13 \quad d=10$$

$$m=3$$

$$n=4$$

$$h \leftarrow 10^2 \bmod 13 = 9$$

$$p \leftarrow 0$$

$$t_0 \leftarrow 0$$

$$i = 1 \text{ to } 3$$

$$p = d \cdot p + P[i] \bmod q$$

$$t_0 = d \cdot t_0 + T[i] \bmod q$$

$$\boxed{i=1}$$

$$p = 10(0) + 4 \bmod 13 = 4$$

$$t_0 = 10(0) + 3 \bmod 13 = 3$$

$$\boxed{i=2}$$

$$p = 10(4) + 2 \bmod 13 = 42 \bmod 13 = 3$$

$$t_0 = 10(3) + 4 \bmod 13 = 34 \bmod 13 = 8$$

$$\boxed{i=3}$$

$$p = 10(3) + 1 \bmod 13 = 31 \bmod 13 = 5$$

$$t_0 = 10(8) + 2 \bmod 13 = 82 \bmod 13 = 4$$

if $8 \leq 0$ to 1
 $8 < m - m$

$$0 < 1$$

$$t_1 = (10(4 - 3 \cdot 9) + 1) \bmod 13$$

$$(d(t_s - T[s+1]R) + T[s+m+1]) \bmod q$$

$$t_1 = (10(4 - 27) + 1) \bmod 13$$

$$= (-230 + 1) \bmod 13$$

$$= (-229 + (13 \times 18)) \bmod 13 = 5 \bmod 13$$

$$a=1$$

$$b=2$$

$$c=3$$

$$d=4$$

$$e=5$$

$$f=6$$

$$g=7$$

$$h=8$$

$$i=9$$

$$j=10$$

Note: In modular arithmetic for finding modulus of -ve number, we can make it positive by adding 13 multiple times.
Since $13 \bmod 13 = 0$.

$$s = 1$$

if $p = 6$, then print "pattern occurs with shift 1"
 $5 = 5$

O/p

Pattern occurs with shift 1.